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22nd Asian Breast Disease Association (ABDA) Annual Scientific Meeting

Strategising Multidisciplinary Management of Breast Diseases

ARE RADIOLOGISTS BETTER THAN STANDALONE ARTIFICIAL INTELLIGENCE FOR MAMMOGRAPHIC BREAST CANCER DETECTION ? : A META-ANALYSIS

Standalone AI • Breast cancer • Mammography • Diagnostic test accuracy • Radiologists

Adamilzary Fikry Abdulmannan[1], Terawan Agus Putranto[4,5], Vininta Fazharyasti[3,5], Nataliandra Natasha Maarifat[2,5]

[1]Radiology Resident, [2]Women's Imaging Radiologist, [3]Thoracic Radiologist, [4]Interventional Radiologist — Department of Radiology, Gatot Soebroto Army Hospital, Jakarta, Indonesia
[5]Department of Radiology, Faculty of Medicine, Universitas Pembangunan Nasional "Veteran" Jakarta, Indonesia



BACKGROUND

Mammography is the cornerstone of breast cancer screening, yet sensitivity is imperfect. A global shortage of breast radiologists and rising screening volumes have driven interest in standalone artificial intelligence (AI) as a first reader or decision-support tool.
Evidence directly comparing standalone AI with radiologists remains inconsistent, and no pooled estimate of their relative accuracy is available.

OBJECTIVES

- **Compare standalone AI and radiologists** for mammographic breast cancer detection.
- Evaluate pooled **diagnostic accuracy (sensitivity, specificity).**
- Assess overall discriminative performance **(DOR, LR+, LR−, SROC).**
- **Define the appropriate clinical role of AI in breast cancer screening.**

METHODS

Systematic review and meta-analysis reported per PRISMA 2020 and PRISMA-DTA.
Five databases searched 2016 to 2026: PubMed, Semantic Scholar, OpenAlex, Google Scholar, Crossref (plus citation-register cross-checking). Included: studies with extractable 2×2 data (**TP, FP, FN, TN) directly comparing standalone AI** with radiologists on mammography **4 studies • 512,247 mammograms • 4,687 confirmed cancers** (Lunit INSIGHT MMG ×3, Transpara ×1; Denmark ×2, South Korea ×2). **Pooling by DerSimonian–Laird random-effects**; **SROC by Moses-Shapiro–Littenberg**; **heterogeneity by Cochran Q and I²; risk of bias by QUADAS-2.**

Study selection — PRISMA 2020 flow

Records identified n = 1,695
5 databases + citation registers, searched to June 2026

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Records screened n = 1,457
238 duplicate records removed

↓

Full-text reports assessed n = 17
1,440 excluded on title/abstract

↓

Studies included n = 4
13 excluded after full-text review

RESULTS

4 studies • 512,247 mammographic examinations • 4,687 confirmed breast cancers

Pooled metric	Standalone AI	Radiologists	p	I² AI / Rad
Sensitivity	0.608 (0.580–0.636)	0.600 (0.556–0.643)	0.753	63.0 / 85.2%
Specificity	0.961 (0.946–0.972)	0.978 (0.977–0.979)	<0.001	99.5 / 67.3%
DOR	41.70 (29.31–59.32)	75.99 (71.15–81.15)	0.001	94.9 / 0.0%
LR+	15.94 (11.47–22.15)	28.40 (27.58–29.25)	<0.001	98.7 / 0.0%
LR−	0.406 (0.379–0.436)	0.379 (0.355–0.404)	0.151	62.4 / 48.7%

Table 1. Pooled diagnostic accuracy (95% CI) of standalone AI vs radiologists — DerSimonian–Laird random-effects model.

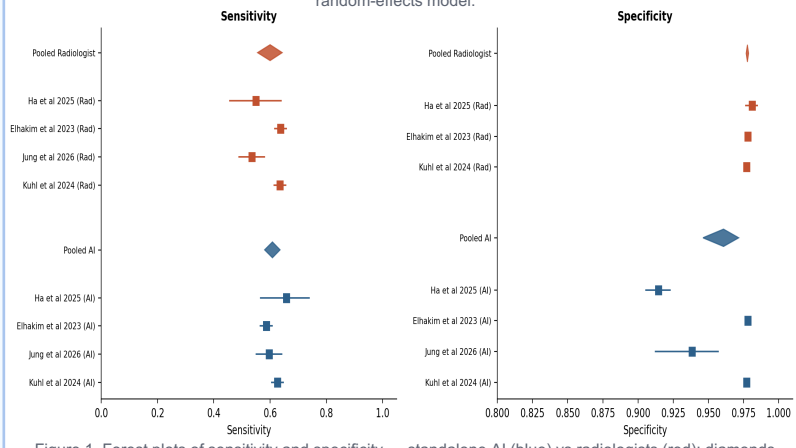


Figure 1. Forest plots of sensitivity and specificity — standalone AI (blue) vs radiologists (red); diamonds = pooled random-effects estimates.

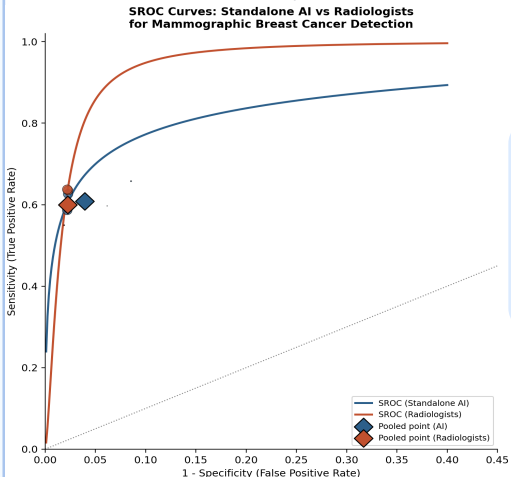


Figure 2. Summary ROC curves for standalone AI (blue) and radiologists (red) — circles = study operating points (size ∝ sample size); diamonds = pooled summary points.

KEY NUMERICAL FINDINGS

Standalone AI vs Radiologists

Sensitivity : 60.8% vs 60.0%
Specificity : 96.1% vs 97.8%
DOR : 41.7 vs 75.9
LR+ : 15.94 vs 28.40
LR- : 0.405 vs 0.379

CONCLUSION

1. Main finding
Standalone AI and radiologists detect a similar proportion of breast cancers **(60.8% vs 60.0%, p = 0.753),** but **radiologists have significantly higher specificity, DOR and LR+.**

2. Clinical meaning
Radiologists currently show superior overall diagnostic performance, achieved through fewer false-positive recalls rather than higher cancer detection.

3. Next step
AI is best positioned as a decision-support / triage tool within double reading; prospective multicentre, multi-vendor studies with pre-specified thresholds are needed.

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ACKNOWLEDGEMENTS

- Department of Radiology, Gatot Soebroto Army Hospital, Jakarta, Indonesia
- Department of Radiology, Faculty of Medicine, Universitas Pembangunan Nasional "Veteran" Jakarta, Indonesia.
- No external funding.
- The authors declare no conflict of interest.
- The review protocol was not prospectively registered in PROSPERO.